

FastAPI Deployment on AWS EC2 using Docker (Step-by-Step Guide)

PART 1 — Create EC2 Instance

Step 1: Login to AWS

Go to: <https://console.aws.amazon.com/>

Search for **EC2**

Step 2: Launch Instance

- Name: `fastapi-server`
 - AMI: Ubuntu (latest)
 - Instance Type: `t2.micro` (Free Tier)
-

Step 3: Create Key Pair

- Name: `fastapi-key`
- Type: RSA
- Format: `.pem`

Download and save securely

Step 4: Configure Security Group

Add inbound rules:

Type	Port	Source
SSH	22	0.0.0.0/0
HTTP	80	0.0.0.0/0
Custom TCP	8000	0.0.0.0/0

Step 5: Launch Instance

Click **Launch Instance**

Step 6: Copy Public IP

Go to **EC2** → **Instances** → Copy **Public IPv4** address

PART 2 — Connect to EC2

Step 7: Set Permission for Key

```
chmod 400 fastapi-key.pem
```

Step 8: Connect via SSH

```
ssh -i fastapi-key.pem ubuntu@YOUR_PUBLIC_IP
```

PART 3 — Setup Server

Step 9: Update System

```
sudo apt update && sudo apt upgrade -y
```

Step 10: Install Docker

```
sudo apt install docker.io -y
```

Step 11: Start Docker

```
sudo systemctl start docker  
sudo systemctl enable docker
```

Step 12: Add User to Docker Group

```
sudo usermod -aG docker ubuntu
```

Then logout and reconnect:

```
exit  
ssh -i fastapi-key.pem ubuntu@YOUR_PUBLIC_IP
```

PART 4 — Upload Project

Step 13: Install Git

```
sudo apt install git -y
```

Step 14: Clone Repository

```
git clone YOUR_GITHUB_REPO_URL  
cd YOUR_PROJECT_FOLDER
```

PART 5 — Docker Setup

Step 15: Verify Dockerfile

```
FROM python:3.10-slim  
  
WORKDIR /app  
  
COPY requirements.txt .  
  
RUN pip install --no-cache-dir -r requirements.txt  
  
COPY . .  
  
CMD ["uvicorn", "main:app", "--host", "0.0.0.0", "--port", "8000"]
```

Step 16: Verify requirements.txt

```
fastapi  
uvicorn  
sqlalchemy  
python-jose  
passlib[bcrypt]
```

PART 6 — Build & Run

Step 17: Build Docker Image

```
docker build -t fastapi-app .
```

Step 18: Run Container

```
docker run -d -p 8000:8000 fastapi-app
```

Step 19: Check Running Containers

```
docker ps
```

PART 7 — Access Application

Step 20: Open in Browser

```
http://YOUR_PUBLIC_IP:8000
```

Step 21: Open Swagger UI

```
http://YOUR_PUBLIC_IP:8000/docs
```

PART 8 — API Testing Steps

Step 22: Register User

POST /api/v1/register

```
{
  "fullname": "Ashu",
  "email": "ashu@gmail.com",
  "password": "123456"
}
```

Step 23: Login User

POST /api/v1/login — Form Data:

```
username = ashu@gmail.com
password = 123456
```

Step 24: Copy Access Token

Response:

```
{
  "access_token": "TOKEN_HERE",
  "token_type": "bearer"
}
```

Step 25: Authorize API

Click **Authorize** and enter:

Bearer TOKEN_HERE

Step 26: Test Protected APIs

Dashboard - GET /api/v1/dashboard

Create Employee - POST /api/v1/employ

```
{
  "fullname": "John",
  "email": "john@gmail.com",
  "isOnProject": true,
  "experience": 3,
  "completed": 5,
  "description": "Backend Dev"
}
```

Get All Employees - GET /api/v1/employs

Get Single Employee - GET /api/v1/employ/{id}

Update Employee - PUT /api/v1/employ/{id}

Delete Employee - DELETE /api/v1/employ/{id}

PART 9 — Debugging

Check Logs

```
docker logs <container_id>
```

Test Locally Inside EC2

```
curl http://localhost:8000
```

PART 10 — SQLite Warning

Current config:

```
sqlite:///./app.db
```

Issue: Data is lost when the container stops.

Fix: Use a Volume

```
docker run -d -p 8000:8000 -v $(pwd):/app fastapi-app
```

PART 11 — Key Concepts

Concept	Description
EC2	Virtual server
Docker	Container runtime
Port Mapping	-p 8000:8000
Security Group	Firewall
Uvicorn	ASGI server

PART 12 — Next Steps (Production)

- ☐ Use Nginx (port 80)
- ☐ Add domain name
- ☐ Enable HTTPS (SSL)
- ☐ Switch SQLite → PostgreSQL
- ☐ Use Docker Compose
- ☐ Setup CI/CD